

Open letter to the authors of Embodied and embedded ecological rationality

To: Dr. Samuel A. Nordli, Dr. Peter M. Todd — co-authors of Embodied and embedded ecological rationality: A common vertebrate mechanism for action selection underlies cognition and heuristic decision-making in humans (Frontiers in Psychology, 2022)

From: Juan Gerardo Castro Sánchez · Nature Intelligence Project · ds.gerardocastro@gmail.com

Date: 10 May 2026

Re: Four engineerable patterns from your 2022 conceptual analysis — including the catalog's first citable formal framework and second canonical pattern

Dear Dr. Nordli and Dr. Todd,

I am writing to introduce the **Nature Intelligence Project** and to explain what your 2022 article Embodied and embedded ecological rationality ([DOI 10.3389/fpsyg.2022.841972](https://doi.org/10.3389/fpsyg.2022.841972)) has contributed to it. Your paper is the source of **8 principles, one new collection of four named patterns**, the catalog's **second canonical pattern** (promoted on cross-substrate evidence), and the catalog's **first citable formal framework**. I want you to know what we extracted, how it sits relative to your argument, and one specific interpretive claim we carry with explicit confidence-flagging.

The problem the project addresses. AI systems today are built almost entirely on one biological template: a simplified caricature of cortical neurons in mammals, scaled up on text. Two structural problems follow. First — as Yann LeCun has argued — these systems learn patterns in language but do not learn how the world works, so they fail at tasks requiring real-world planning. Second, the biological inspiration has been only one species deep. The diversity of cognitive design across the tree of life — birds, bacteria, cephalopods, slime molds, and the half-billion-year-old vertebrate substrate your paper anchors — is treated as curiosity rather than design library. The project's bet is that the most consequential next-generation architectures need an engineering-facing translation from comparative cognition and neuroscience to reach the people who build AI systems.

The catalog: NEXI. Each entry pairs a **natural exemplar** with an **architectural primitive** and a **drop-in skill specification**. Every entry includes a **falsifiable hypothesis**. Patterns aggregate into **collections** — coherent intelligence models whose member patterns reinforce each other.

What I extracted from your work. Your 2022 conceptual analysis surfaces four architectural primitives. Together they form one collection — Embodied Action Selection — describing the same architectural function under different domain transformations. This isomorphism shape is the catalog's fourth collection shape, distinct from the constellation, pipeline, and composition shapes the prior three sources had produced.

- **Action Selection as Common Substrate.** Drawn from your argument that the CBGTC loop performs the same role in lamprey brains as in modern human brains — virtually unchanged across all living vertebrate species over half a billion years. This is the catalog's **second canonical pattern** (alongside **niche-specification**). Its promotion was justified by **cross-substrate triangulation**: the same architectural function (default-inhibit / selective-disinhibit gating, goal-directed sequence construction, contextual reinforcement) is documented in vertebrate-CBGTC loops (your work), bacterial molecular regulatory networks (Nesin & Chandrankunnel 2025), and cnidarian nerve nets via Turner et al. 2026's phylogenetic placement of jellyfish and sea anemones. Three substrates; one architectural pattern; conservation across the deepest divergences in animal evolution. The engineering translation: AI architectures should treat action selection as a first-class architectural commitment — inspectable, lesionable, shareable across domains — rather than as an emergent property of an opaque substrate.
- **Ecological Context Model (formalism layer).** Drawn directly from §4: the symbolic framework with three numbered equations ($C = \{A(g), E\}$; $B(C) = f(Br, Ba(g, P))$; $g+(C) = f(B, E)$) generalising Lewin's 1936 field-theory equation, plus the §5 state-merger commitment that agent-internal

features are part of the environment E. This is the **catalog's first named formal framework**: prior sources contributed observations the project synthesised after the fact; your paper contributes equations the catalog can adopt directly. The engineering translation: AI systems committing to the formalism gain decision-provenance auditability and transfer-failure attribution that distinguishes perception failure from environment-unobservability failure.

- **Exaptation Architectural Reuse.** Drawn from your master claim (§7): the same machinery vertebrates use for sequential goal-directed motor control was redeployed in human evolution to regulate cognitive control. The internal-external search homology you cite (Hills et al. 2008, 2015a, 2015b; Todd & Hills 2020; Todd & Miller 2018) extends the case to a second exaptive instance. The engineering translation: AI architectures should treat **architectural reuse via exaptation** as a first-class deployment-time design pressure — one well-developed mechanism redeployed across domains via thin per-domain adapters, with cross-domain transfer of improvements as the architectural payoff. The perceptron lineage has overwhelmingly favoured the opposite (specialisation per task).
- **Heuristics as Habits Fusion.** Drawn from your formal equivalence in §6 and §7, including the recognition-heuristic worked example (Tokyo / Yokohama / Nagasaki, citing Goldstein & Gigerenzer 2002). The engineering translation: AI architectures that currently implement Type 1 / Type 2 reasoning as separate modules can unify them under a single context-overlap-driven selection mechanism drawing from a shared repertoire spanning motor and cognitive operations — heuristic failures and motor-habit failures gain the same diagnostic structure, and remedies transfer across domains.

What we deliberately flag for confidence-grading. Your heuristics-as-habits formal equivalence is the catalog's most consequential interpretive move from this source — and we have graded it **core × medium** rather than **core × high** to flag this transparently. **_Core_** because it load-bears the exaptation-of-CBGTC-for-cognition argument. **_Medium_** because the equivalence is interpretive synthesis rather than fresh empirical demonstration — a careful reader could accept every neuroscience point and still reject the formal mapping between recognition-heuristic errors and motor-habit errors. The grading is an **_audit hook_** rather than a critique: a corroborating source (e.g. CBGTC fMRI activation during structurally-induced heuristic errors) would justify a re-grade to **high**. We mention this transparently because honest grading is itself a methodological commitment, and we did not want quiet confidence-marking on what is, in our reading, the paper's sharpest interpretive contribution. The single-goal simplification you note in footnote 6 is logged as an open extension point in the collection's complementarity notes.

An invitation. I would value any feedback you are willing to offer: disagreements with my reading, stronger formulations of the falsifiable claims, additional caveats, primary sources I should cite alongside your synthesis, or pointers from the broader Hills / Todd / Miller research programme that would strengthen the exaptation collection's evidence base. The catalog repository is currently private during initial population; once the threshold for public release is reached, it will live at github.com/Ds-GerardoCastro/NEXI. In the interim I am happy to share the relevant entries directly — please reply to this email.

With genuine appreciation for the structural clarity of your formal model, the depth of the phylogenetic argument, and the architectural ambition of the exaptation thesis — which has given the catalog its first citable formal framework, its second canonical pattern, and a fourth collection shape.

Sincerely,

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